

Angle between Two Lines

Eliminating n as above we have

$$(2l+m)(l-m)=0$$

$$\text{When } 2l+m=0 \text{ then } \frac{l}{1} = \frac{m}{-2} = \frac{n}{1}$$

$$\text{When } l-m=0 \text{ then } \frac{l}{1} = \frac{m}{1} = \frac{n}{-2}$$

 $\therefore$ D.R.'s are 1, -2, 1 and 1, 1, -2
If θ be the angle between the lines then

$$\cos \theta = \frac{\sum a_1 a_2}{\sqrt{\sum a_1^2} \cdot \sqrt{\sum a_2^2}}$$

$$\text{or } \cos \theta = \frac{1 \cdot 1 - 2 \cdot 1 + 1 \cdot (-2)}{\sqrt{6} \cdot \sqrt{6}} = \frac{-3}{6} = -\frac{1}{2}$$

$$= -\cos \pi/3 = \cos (\pi - \pi/3)$$

$$\text{Hence } \theta = \frac{2\pi}{3} = 120^\circ$$

(d) If points P and Q are $(2, -6)$ and $(3, -4, 5)$ find the angle that OP makes with OQ [Rohilkhand 82]

$$\cos \theta = \frac{-18\sqrt{2}}{35}$$

(e) The direction-cosines of two lines are governed by the relations $l+m+n=0$ and $l^2+m^2-n^2=0$. Find the angle between them.

[Raj. 83; Calcut 81]

100% ✓ 17. (a) Prove that the two lines whose direction-cosines are connected by the relations $2l+2m-n=0$ and $lm+mn+nl=0$ are perpendicular to each other.

Proceeding as above and eliminating n , we get

$$2l^2+5lm+2m^2=0 \text{ or } (2l+m)(l+2m)=0.$$

When $2l+m=0$, then from $2l+2m-n=0$, we get $m=n$.

$$\therefore \frac{l}{1} = \frac{m}{-2} = \frac{n}{-2}, \text{ i.e., d.c.'s are proportional to } 1, -2, -2$$

When $l+2m=0$, then from $2l+2m-n=0$, we get $l=n$.

$$\therefore \frac{l}{-2} = \frac{m}{1} = \frac{n}{-2} \text{ i.e., d.c.'s are proportional to } -2, 1, -2.$$

The lines will be perpendicular if $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

$$\text{or } 1(-2) - 2(1) - 2(-2) = 0 \text{ which is true.}$$

100% ✓ 18. (b) Show that the lines whose direction-cosines are connected by $l+m+n=0$ and $2mn+3nl-5lm=0$ are perpendicular to each other. [Meerut 83; Magadh 75; Kurkshetra 74]

$$n = -(l+m) \text{ and } n(2m+3l) - 5lm = 0$$

$$\text{or } -(l+m)(2m+3l) - 5lm = 0 \text{ or } 3l^2 + 10lm + 2m^2 = 0$$

$$\text{or } 3(l/m)^2 + 10(l/m) + 2 = 0$$

If its roots be $\frac{l_1}{m_1}$ and $\frac{l_2}{m_2}$, then

$$\frac{l_1}{m_1} + \frac{l_2}{m_2} = \frac{2}{3} \text{ or } \frac{l_1 l_2}{2} = \frac{m_1 m_2}{3} = k, \text{ say} \quad \dots(1)$$

Again $l = -(m+n)$ and $2mn + l(3n-5m) = 0$
or $2mn - (m+n)(3n-5m) = 0$
or $5m^2 + 4mn - 3n^2 = 0$
or $5\left(\frac{m}{n}\right)^2 + 4\left(\frac{m}{n}\right) - 3 = 0$

If its roots be $\frac{m_1}{n_1}$ and $\frac{m_2}{n_2}$ then

$$\frac{m_1 m_2}{n_1 n_2} = \frac{-3}{5} \text{ or } \frac{m_1 m_2}{3} = \frac{n_1 n_2}{-5} = k, \text{ by (1)} \quad \dots(2)$$

Hence from (1) and (2) we have

$$\frac{l_1 l_2}{2} = \frac{m_1 m_2}{3} = \frac{n_1 n_2}{-5} = k.$$

$$\therefore l_1 l_2 + m_1 m_2 + n_1 n_2 = k(2+3-5) = k \cdot 0 = 0$$

Hence the lines are perpendicular.

(c) If points P, Q are $(2, 3, 4)$ and $(1, -2, 1)$ respectively then prove that OP is perpendicular to OQ where O is origin

[Magadh 68]

18. Prove that the straight lines whose direction-cosines are given by the relations.

$$al + bm + cn = 0 \text{ and } fm + gn + hl = 0$$

are perpendicular if $\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$ and parallel if

$$\sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0$$

or $a^2 f^2 + b^2 g^2 + c^2 h^2 - 2bcgh - 2cahf - 2abfg = 0$

[Allahabad 82, Meerut 84, 82, Sagar 71, 73; Kanpur 73, Rohailkhand 81; Indore 81]

Eliminating l between the given relations, we get

$$fm + (gn + hm) \left(\frac{bm + cn}{-a} \right) = 0 \quad \therefore l = \left(\frac{bm + cn}{-a} \right)$$

or $bhm^2 + mn(ch + bg - af) + cgn^2 = 0$

or $bh\left(\frac{m}{n}\right)^2 + (ch + bg - af)\frac{m}{n} + cg = 0$

Let $\frac{m_1}{n_1}$ and $\frac{m_2}{n_2}$ be the roots of the above equation; then in case the lines be parallel, their direction-cosines are equal and

hence, both the roots of (1) should be equal the condition for which is $B^2 - 4AC = 0$.

$$\text{i.e.} \quad (ch + bg - af)^2 = 4bcgh \quad \dots(A)$$

$$\text{or} \quad ch + bg - af = \pm 2\sqrt{bcgh}$$

$$\text{or} \quad ch + bg \pm 2\sqrt{bcgh} = af$$

$$\text{or} \quad [\sqrt{ch} \pm \sqrt{bg}]^2 = [\sqrt{af}]^2$$

$$\text{or} \quad \pm[\sqrt{ch} \pm \sqrt{bg}] = \sqrt{af}$$

$$\text{or} \quad \sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0 \text{ is the required condition.}$$

The second form of condition is obtained by simplifying result (A).

Again from (1) we have $\frac{m_1}{n_1} \cdot \frac{m_2}{n_2} = \text{product of the roots of (1)}$.

$$\text{or} \quad \frac{m_1 m_2}{n_1 n_2} = \frac{cg}{bh}; \quad \therefore \frac{m_1 m_2}{cg} = \frac{n_1 n_2}{bh}$$

$$\text{or} \quad \frac{m_1 m_2}{g/b} = \frac{n_1 n_2}{h/c} \quad \dots(1)$$

Similarly if we eliminate n between the given relations, we shall have

$$\frac{l_1 l_2}{f/a} = \frac{m_1 m_2}{g/b} \quad \dots(2)$$

$$\therefore \frac{l_1 l_2}{f/a} = \frac{m_1 m_2}{g/b} = \frac{n_1 n_2}{h/c} \text{ from (2) and (3)}$$

Now if the lines are perpendicular, then

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

$$\therefore \frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0 \text{ is the required condition.}$$

19. Prove that the two lines whose direction cosines are connected by the two relations $al + bm + cn = 0$ and $ul^2 + vm^2 + wn^2 = 0$ are perpendicular if

$$a^2(v+w) + b^2(w+u) + c^2(u+v) = 0$$

$$\text{and parallel if } \frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0.$$

[Agra 74, Indore 74, Kanpur 78, Raj. 78, Meerut 84, Punjab 73]

Eliminating n between the given relations, we get

$$ul^2 + vm^2 + w \left(\frac{al + bm}{-c} \right)^2 = 0$$

$$\text{or} \quad (c^2 u + a^2 w) \frac{l^2}{m^2} + 2abw \cdot \frac{l}{m} + (b^2 w + c^2 v) = 0 \quad \dots(1)$$

$$\therefore \frac{l_1}{m_1} \cdot \frac{l_2}{m_2} = \text{product of roots} = \frac{b^2 w + c^2 v}{c^2 u + a^2 w}$$

or $\frac{l_1 l_2}{b^2 w + c^2 v} = \frac{m_1 m_2}{c^2 u + a^2 w} = \frac{n_1 n_2}{a^2 v + b^2 u}$ by symmetry.

If the lines be perpendicular, then $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$
 or $a^2(v+w) + b^2(w+u) + c^2(u+v) = 0$.

Proved

Again if the lines be parallel, then their d.c.'s are equal so that the roots of (1) should be equal, i.e., $B^2 - 4AC = 0$

or $4a^2 b^2 w^2 - 4(c^2 u + a^2 w)(b^2 w + c^2 v) = 0$

or $a^2 c^2 v w + b^2 c^2 u w + c^4 u v = 0$

or $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$ on dividing by $c^2 u v w$.

3

The Plane

§ 1. To prove that the general equation of first degree in x, y, z i.e., $Ax + By + Cz + D = 0$ represents a plane.

[Vikram 75; Madras 71; Pb 82; Bihar 76; Kanpur 75; Ranchi 74]

Corresponding theorem of plane geometry.

General equation of first degree in x, y i.e., $Ax + By + C = 0$ represents a straight line.

Definition of a plane. A plane is a surface such that if any two points are taken on it, the straight line joining them lies wholly in the surface i.e. every point on the line joining the two points will be on the plane.

From the above definition of a plane we shall prove our result. Let (x_1, y_1, z_1) and (x_2, y_2, z_2) be the co-ordinates of any two points lying on the locus whose equation is

$$Ax + By + Cz + D = 0 \quad \dots(1)$$

$$\therefore Ax_1 + By_1 + Cz_1 + D = 0 \quad \dots(2)$$

$$Ax_2 + By_2 + Cz_2 + D = 0 \quad \dots(3)$$

Multiplying (3) by λ and adding to (2), we get

$$A(x_1 + \lambda x_2) + B(y_1 + \lambda y_2) + C(z_1 + \lambda z_2) + D(1 + \lambda) = 0$$

$$\text{or } \frac{A(x_1 + \lambda x_2)}{(1 + \lambda)} + \frac{B(y_1 + \lambda y_2)}{(1 + \lambda)} + \frac{C(z_1 + \lambda z_2)}{(1 + \lambda)} + D = 0. \quad \dots(4)$$

Result (4) shows that equation (1) is satisfied by the point whose co-ordinates are

$$\left(\frac{x_1 + \lambda x_2}{1 + \lambda}, \frac{y_1 + \lambda y_2}{1 + \lambda}, \frac{z_1 + \lambda z_2}{1 + \lambda} \right).$$

But from § 4 Cor. 2, P. 10 we know that these are the General Co-ordinates of a point on the line joining the points (x_1, y_1, z_1) and (x_2, y_2, z_2) .

Hence we conclude that if (1) is satisfied by any two points then every point on the line joining these two also satisfies it and therefore it represents a plane.

Cor 1. General equation of a plane that passes through a given point.

Let the equation of the plane be $Ax + By + Cz + D = 0$. Since it passes through (x_1, y_1, z_1) we have

$$Ax_1 + By_1 + Cz_1 + D = 0.$$

Subtracting, we get $A(x - x_1) + B(y - y_1) + C(z - z_1) = 0$.

Cor 2. General equation of a plane through origin.

$Ax + By + Cz = 0$ as $D = 0$ $\therefore$ it passes through origin.

Number of arbitrary constants in the general equation of a plane.

$$Ax + By + Cz + D = 0 \text{ or } \frac{A}{D}x + \frac{B}{D}y + \frac{C}{D}z + 1 = 0.$$

The above form shows that there are three arbitrary constants

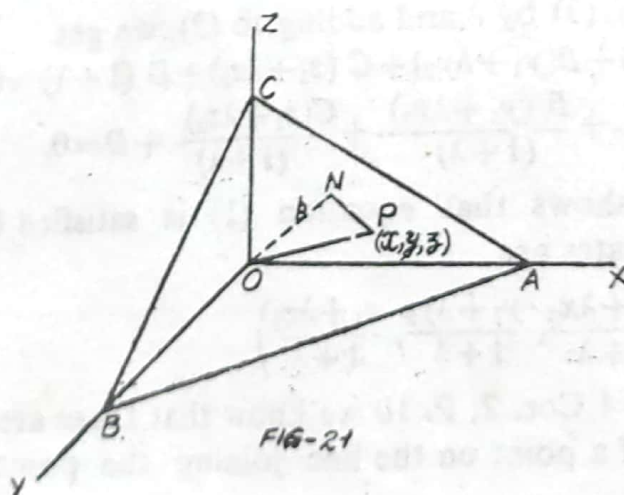
$\frac{A}{D}, \frac{B}{D}, \frac{C}{D}$ occurring in the equation of the plane. Hence a plane can be found to satisfy three conditions, each condition giving us the value of a constant. Thus a plane can be made to pass through three non-collinear points, or can pass through two given points and perpendicular to a given plane, or pass through a given point and perpendicular to two given planes etc., etc.

§ 2. Normal form of the equation of a plane. To find the equation of a plane in terms of p the perpendicular from origin on the plane and direction cosines l, m, n , of this perpendicular.

(Venka 84; Gauhati 74; Bhopal 71)

Corresponding form in Plane Geometry.

The equation of a straight line in terms of p the length of the



perpendicular to it from origin and angle α which this perpendicular makes with positive direction of x-axis i.e.,

$$x \cos \alpha + y \sin \alpha = p.$$

Let $ON=p$ be the perpendicular from origin O to the plane ABC , N being the foot of the perpendicular and the direction-cosines of ON are l, m, n .

Now take P any point (x, y, z) on the plane. Since ON is perpendicular to the plane, it is perpendicular to every line lying in the plane. Therefore PN is perpendicular to ON and hence by definition $ON=p$ is the projection of the line OP on the line ON .

Now we know from § 8, P. 21, that the projection of a line joining (x_1, y_1, z_1) and (x_2, y_2, z_2) on another line whose direction-cosines are l, m, n is

$$l(x_1 - x_2) + m(y_1 - y_2) + n(z_1 - z_2).$$

Hence the projection of OP on ON is

$$l(x-0) + m(y-0) + n(z-0) = ON$$

or $lx + my + nz = p$ is the required equation of the plane, it being a relation between the co-ordinates of any point (x, y, z) on the plane.

Note. p is always positive.

§ 8. Intercepts form. To find the equation to the plane in terms of the intercepts a, b and c which it makes on the axis.

(Bhagalpur 75; Venkateshwara 74; Madras 73 ;
Sagar 71 ; Ranchi 80 ; Magadh 76; Indore 69)

Let the equation of the plane be

$$Ax + By + Cz + D = 0.$$

Since it makes intercepts a, b and c on the axes of x, y and z respectively, then the points $(a, 0, 0), (0, b, 0), (0, 0, c)$ will satisfy (1).

$$\therefore Aa + D = 0, Bb + D = 0, Cc + D = 0.$$

$$\therefore A = -\frac{D}{a}, B = -\frac{D}{b}, C = -\frac{D}{c}$$

Putting the values of A, B and C in (1), we get

$$-\frac{D}{a}x - \frac{D}{b}y - \frac{D}{c}z + D = 0 \quad \text{or} \quad \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

is the required equation of the plane.

Corresponding form of plane geometry :

The equation of a straight line in terms of the intercepts a and b it makes on the axes of x and y respectively is

$$x/a + y/b = 1.$$

Ex. 1. A plane meets the co-ordinate axes in A, B, C such that the centroid of the triangle ABC is the point (p, q, r) show that the equation of the plane is

$$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 3. \quad (\text{Burdwan 1976; Kanpur 81; Madras 73; Venkateshwara 74; Ranchi 79; Magadh 77})$$

Let the equation of the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ (1)

$\therefore A$ is $(a, 0, 0)$, B is $(0, b, 0)$ and C is $(0, 0, c)$.

Centroid of the triangle ABC is

$$\frac{a+0+0}{3}, \frac{0+b+0}{3}, \frac{0+0+c}{3} \quad [\S 5 P. 9]$$

i.e. $(\frac{a}{3}, \frac{b}{3}, \frac{c}{3})$. But we are given that the centroid is the point

$$(p, q, r), \therefore \frac{a}{3} = p, \frac{b}{3} = q, \frac{c}{3} = r.$$

or $a = 3p$, $b = 3q$ and $c = 3r$. Putting the values of a , b and c in (1), we get the required equation of the plane as

$$\frac{x}{3p} + \frac{y}{3q} + \frac{z}{3r} = 1 \text{ or } \frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 3.$$

~~§ 4. Equation of a plane passing through three given points.~~

(Rewa 74; Agra 82; Indore 76; Gorakhpur 77; Magadh 77; Bhagalpur 83; Madras 83)

To find the equation of a plane passing through three points

(x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) .

Let the equation of the plane be

$$Ax + By + Cz + D = 0 \quad \dots (1)$$

Since it passes through the three given points, we have

$$Ax_1 + By_1 + Cz_1 + D = 0, \quad \dots (2)$$

$$Ax_2 + By_2 + Cz_2 + D = 0. \quad \dots (3)$$

$$Ax_3 + By_3 + Cz_3 + D = 0. \quad \dots (4)$$

Eliminating A , B , C and D from (1), (2), (3) and (4), we get the equation of the plane

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0. \quad \dots (6)$$

Condition for four points to lie in a plane.

Let the fourth point be (x_4, y_4, z_4) . If it lies on the plane that passes through the other three points, then its co-ordinates should satisfy (5). Hence the required condition is

*

$$\begin{vmatrix} x_4 & y_4 & z_4 & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0 \text{ or } \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0.$$

Note: In numerical examples it will be difficult for the students to expand the above determinant of fourth order to find the equation to the plane passing through three given points. The method given below may be adopted.

Ex. 2. (a) Show that the four points $(0, 4, 3)$, $(-1, -5, -3)$, $(-2, -2, 1)$ and $(1, 1, -1)$ are coplanar, and find the common plane.

Here we shall find the equation of the plane passing through any three points say $(0, 4, 3)$, $(-2, -2, 1)$ and $(1, 1, -1)$ and then show that the fourth point $(-1, -5, -3)$, satisfies it.

General equation of a plane through the point $(0, 4, 3)$ is

$$A(x-0) + B(y-4) + C(z-3) = 0 \quad \dots(1)$$

[P. 38. Cor. 1]

It will pass through $(-2, -2, 1)$ and $(1, 1, -1)$ if

$$-2A - 6B - 2C = 0 \text{ or } A + 3B + C = 0 \quad \dots(2)$$

and $A - 3B - 4C = 0 \text{ or } A - 3B - 4C = 0 \quad \dots(3)$

Solving (2) and (3) by the method of cross-multiplication we

get $\frac{A}{-12+3} = \frac{B}{1+4} = \frac{C}{-3-3}$ or $\frac{A}{-9} = \frac{B}{5} = \frac{C}{-6} = k$ say:

$$\therefore A = -9k, B = 5k \text{ and } C = -6k.$$

Putting the values of A , B and C in (1) and cancelling k , we get the equation of the plane as

$$-9(x-0) + 5(y-4) - 6(z-3) = 0$$

or $9x - 5y + 6z + 2 = 0$

and clearly the co-ordinates of the point $(-1, -5, -3)$ satisfy it and hence the given four points are coplanar and equation of the common plane is $9x - 5y + 6z + 2 = 0$.

(b) Show that the four points $(0, -1, -1)$, $(4, 5, 1)$, $(3, 9, 4)$ and $(-4, 4, 4)$ lie in a plane. *conv.*

(Sagar 71; Agra 60; Meerut 82)

The equation of the plane in which they lie is

$$5x - 7y + 11z + 4 = 0$$

(c) Prove that the four points whose co-ordinates are $(0, -1, 0)$, $(2, 1, -1)$, $(1, 1, 1)$ and $(3, 3, 0)$ are coplanar.
(Agra 78, 76 ; Utkal 78; Rewa 74 ; Jiwaji 60)

§ 5. Reduction of the general equation of the plane to
(i) Intercepts form, (ii) Normal form.

$$Ax + By + Cz + D = 0.$$

Transpose D on the other side and then divide through-out by $-D$ so as to make the R.H.S. = 1.

$$(i) Ax + By + Cz = -D.$$

$$\text{or } \frac{A}{-D}x + \frac{B}{-D}y + \frac{C}{-D}z = 1 \text{ or } \frac{x}{-D/A} + \frac{y}{-D/B} + \frac{z}{-D/C} = 1$$

which is of the form $x/a + y/b + z/c = 1$.

Hence the intercept on the axis of x is $-D/A$.

i.e. $\frac{\text{constant term}}{\text{coefficient of } x}$ with sign changed.

Similarly the intercepts on the axes of y and z are found.

Ex. 3 What are the intercepts of the plane.

$$4x + 3y - 12z + 6 = 0$$

on the axes ?

Ans. $-\frac{3}{2}$, -2 , $\frac{1}{2}$ respectively.

(ii) Normal Form

If the general equation of the plane

$$Ax + By + Cz + D = 0$$

is of the same form as the normal form $lx + my + nz - p = 0$, then on comparison, we have

$$\frac{l}{A} = \frac{m}{B} = \frac{n}{C} = -\frac{p}{D}$$

$$\text{or } l = -\frac{A}{D}p, m = -\frac{B}{D}p, n = -\frac{C}{D}p. \quad \dots(1)$$

Since l, m, n are the direction-cosines of the normal to the plane, we have $l^2 + m^2 + n^2 = 1$;

$$\therefore \frac{A^2 + B^2 + C^2}{D^2} p^2 = 1 \text{ or } p = \pm \frac{D}{\sqrt{A^2 + B^2 + C^2}} \quad \dots(2)$$

Now we always keep p positive. Hence if D be -ive then from (2), we must choose $p = -\frac{D}{\sqrt{A^2 + B^2 + C^2}}$ and hence from (1).

$$l = + \frac{A}{\sqrt{A^2+B^2+C^2}}, m = + \frac{B}{\sqrt{A^2+B^2+C^2}},$$

$$n = + \frac{C}{\sqrt{A^2+B^2+C^2}}$$

Putting for l, m, n and p in $lx+my+nz=p$, we get

$$\frac{A}{\sqrt{A^2+B^2+C^2}} x + \frac{B}{\sqrt{A^2+B^2+C^2}} y + \frac{C}{\sqrt{A^2+B^2+C^2}} z$$

$$= - \frac{D}{\sqrt{A^2+B^2+C^2}} \quad [D - \text{ive}]$$

Again if D be +ive, then from (2), we must choose

$$p = \frac{D}{\sqrt{A^2+B^2+C^2}} \text{ and hence from (1) } l = - \frac{A}{\sqrt{A^2+B^2+C^2}}$$

$$m = - \frac{B}{\sqrt{A^2+B^2+C^2}}, n = - \frac{C}{\sqrt{A^2+B^2+C^2}}$$

Putting these values of l, m, n and p in $lx+my+nz=p$, we get

$$- \frac{A}{\sqrt{A^2+B^2+C^2}} x - \frac{B}{\sqrt{A^2+B^2+C^2}} y - \frac{C}{\sqrt{A^2+B^2+C^2}} z$$

$$= \frac{D}{\sqrt{A^2+B^2+C^2}} \quad [D + \text{ive}]$$

Working rule :— Divide the given equation throughout by

$\sqrt{A^2+B^2+C^2}$ i.e. by

$$\sqrt{[(\text{coeff. of } x)^2 + (\text{coeff. of } y)^2 + (\text{coeff. of } z)^2]}.$$

Transpose the constant term to the R. H. S. and see that it is positive. If it be not positive, then change the sign throughout so that it becomes positive.

Direction-cosine of normal to a plane : From (1) we find that the direction-cosines of the normal to the plane represented by $Ax+By+Cz+D=0$ are proportional to A, B and C . In other words, the d.c.'s are proportional to the coefficients of x, y and z .

Ex. 4. Reduce the equation of the plane $2x-3y+6z=7$ to the normal form and hence find the direction cosines of the normal and its length.

Dividing throughout by $\sqrt{2^2+(-3)^2+(6)^2}$ i.e. by 7, the equation of the plane takes the form

$$\frac{2}{7}x - \frac{3}{7}y + \frac{6}{7}z = \frac{7}{7}. \quad \dots(1)$$

The constant term in the R. H.S. is +ive. If it were negative, then we would have changed the sign throughout to make it +ive.

Now the equation (1) is of the form $lx+my+nz=p$ and hence the direction-cosines of the normal are $(\frac{2}{7}, -\frac{3}{7}, \frac{6}{7})$ and length of the normal is 1.

Ex. 5. O is the origin and A is the point (a, b, c) . Find the direction-cosines of the join of OA and deduce the equation of the plane through A at right angles to OA .

(Rohilkhand 83; Dibrugarh 71; Indore 62; Patna 60)

D.R.'s of OA are $a=0, b=0, c=0$ or a, b, c .

$\therefore$ D.C.'s of OA are

$$\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$$

Equation of any plane through A , i.e. (a, b, c) is

$$A(x-a) + B(y-b) + C(z-c) = 0. \quad (\text{Cor. 1 P. 38})$$

But plane is at right angles to OA which therefore is normal and whose D.R.'s are a, b, c .

Hence the plane is $a(x-a) + b(y-b) + c(z-c) = 0$

or $ax + by + cz = a^2 + b^2 + c^2,$

§ 6. Angle between two planes. By angle between two planes we mean angle between normals to the planes drawn from any point.
[Magadh 75; Nagpur 58]

Let the two planes be

$$A_1x + B_1y + C_1z + D_1 = 0 \text{ and } A_2x + B_2y + C_2z + D_2 = 0.$$

The direction-ratios of normals to the two planes are A_1, B_1, C_1 and A_2, B_2, C_2 . Hence if θ be the angle between them, then

$$\cos \theta = \frac{A_1A_2 + B_1B_2 + C_1C_2}{\sqrt{A_1^2 + B_1^2 + C_1^2} \sqrt{A_2^2 + B_2^2 + C_2^2}} \quad \dots (4)$$

Conditions for the two planes to be parallel or perpendicular.

If two planes are perpendicular to each other so are their normals whose direction-cosines are proportional to A_1, B_1, C_1 and A_2, B_2, C_2 . Hence the condition is

$$A_1A_2 + B_1B_2 + C_1C_2 = 0. \quad [\text{P. 24}]$$

i.e. sum of the products of the coefficients of x, y and z is zero.

Similarly if two planes are parallel, so are their normals and hence the condition is that

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2}, \quad [\text{r. 24}]$$

i.e. the direction-ratios which are given by the coefficients of x, y and z in two equations are proportional.

From above we can easily conclude that the equations of two parallel planes differ only in the constant term.

Note. The above result is identical with the corresponding result of plane geometry.

Ex. 6. Find the angle between the planes

$2x - y + z = 6$ and $x + y + 2z = 7$.
If θ be the angle between the planes, then

$$\cos \theta = \frac{2 \cdot 1 + (-1) \cdot 1 + 1 \cdot 2}{\sqrt{2^2 + (-1)^2 + 1^2} \sqrt{1^2 + 1^2 + 2^2}} = \frac{3}{\sqrt{6} \sqrt{6}} = \frac{1}{2};$$

$$\therefore \theta = \pi/3.$$

Ex. 7. Find the equation of the plane through the point $(4, 0, 1)$ and parallel to the plane $4x + 3y - 12z + 6 = 0$.

We know that equations of the parallel planes differ only in constant term and hence the equation of the plane parallel to $4x + 3y - 12z + 6 = 0$ is

$$4x + 3y - 12z + k = 0.$$

Since it passes through $(4, 0, 1)$, we get

$$16 + 0 - 12 + k = 0 \text{ or } k = -4.$$

Hence the required equation of the plane is

$$4x + 3y - 12z - 4 = 0.$$

§ 7. General equation of the plane through the intersection of two given planes. [Indore 84; Jiawaji 66]

Let the two planes be given by

$$P \equiv A_1x + B_1y + C_1z + D_1 = 0 \quad \dots(1)$$

and

$$Q \equiv A_2x + B_2y + C_2z + D_2 = 0$$

then

$$P + \lambda Q = 0 \quad \dots(2)$$

$$\text{or } (A_1x + B_1y + C_1z + D_1) + \lambda (A_2x + B_2y + C_2z + D_2) = 0. \quad \dots(3)$$

is the required plane through the intersection of (1) and (2) because if any point (x_1, y_1, z_1) satisfies both (1) and (2), then for all values of λ it satisfies (3) also.

Similarly all other points which are common to (1) and (2), i.e. those which lie on the line of intersection of (1) and (2) will lie on (3), and the equation being of 1st degree in x, y and z therefore represents the required plane.

Ex. 8. (a) Find the equation of the plane through the intersection of the planes $x + 2y + 3z - 4 = 0$ and $2x + y - z + 5 = 0$ and perpendicular to the plane $5x + 3y + 6z + 8 = 0$.

[Avadh 82; Luck. 82; Pb. 76; Allahabad 79]

Any plane through the intersection of the given planes is

$$(x + 2y + 3z - 4) + \lambda (2x + y - z + 5) = 0$$

or

$$x(1 + 2\lambda) + y(2 + \lambda) + z(3 - \lambda) - (4 - 5\lambda) = 0 \quad \dots(1)$$

Since the plane (1) is perpendicular to the plane

$$5x + 3y + 6z + 8 = 0$$

$$\therefore 5(1 + 2\lambda) + 3(2 + \lambda) + 6(3 - \lambda) = 0 \quad [\S 6 P. 44]$$

or

$$29 + 7\lambda = 0; \lambda = -29/7.$$

Putting the value of λ in (1) we get the eqn of the required plane as

$$-51x - 15y + 50z - 173 = 0 \text{ or } 51x + 15y - 50z + 173 = 0$$

Putting the value of λ in (1), we get the equation of the required plane as

$$-51x - 15y + 50z - 173 = 0 \text{ or } 51x + 15y - 50z + 173 = 0$$

Qb) Find the equation of the plane through the intersection of planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to plane $4x + 5y - 3z = 8$.

[Meerut 81; Kanpur 78]

Ans. $28x - 17y + 9z = 0$.

§ 8. Condition that a line whose direction-ratios are p, q, r may be parallel to a plane or be perpendicular to the given plane. What do you mean by the angle between a line and a plane?

Let the plane be $Ax + By + Cz + D = 0$.

If the line with direction-ratios p, q, r is parallel to the plane, then the normal to the plane whose direction-ratios are A, B, C is perpendicular to the line and hence the required condition is

$$Ap + Bq + Cr = 0. \quad [P. 24]$$

Similarly if the line be perpendicular to the plane then it is parallel to the normal of the plane. Hence the required condition is

$$\frac{A}{p} = \frac{B}{q} = \frac{C}{r} \quad [P. 24]$$

Angle between a line and a plane.

By angle between a line and a plane we mean complement of the angle between a line and normal to the plane.

§ 9. General equation of a plane in some particular cases.

1. Equation to Co-ordinate Planes.

We know that every point in the yz -plane will have its x -co-ordinate zero and hence its equation is $x=0$. Similarly the equations of the zx and xy -planes are $y=0$ and $z=0$ respectively.

2. Planes parallel to Co-ordinate planes.

If we draw a plane parallel to the yz -plane at a distance a from it, then every point in the plane will have its x -co-ordinate equal to a and have its equation as $x=a$.

Similarly equation of the plane parallel to zx -plane is $y = \text{const.}$ and the plane parallel to xy -plane is $z = \text{constant.}$

3. Planes perpendicular to Co-ordinate axis.

A plane which is perpendicular to x -axis is clearly parallel to yz -plane and hence by the equation of the planes which are perpendicular to axis we mean the planes which are parallel to the co-ordinate planes. [See Case 2]

4. Planes perpendicular to co-ordinate planes. [Indore 82]

Now yz -plane is $x=0$ or $x+0y+0z=0$... (1)

Let the equation to the plane perpendicular to yz -plane be

$$ax+by+cz+d=0 \quad \dots (2)$$

The planes (1) and (2) are perpendicular

$$\therefore 1.a+0.b+0.c=0 \quad \text{or} \quad a=0$$

Hence the equation of the plane perpendicular to yz -plane is

$$by+cz+d=0$$

Similarly the equation of the planes perpendicular to zx and xy planes are

$$ax+cz+d=0 \quad \text{and} \quad ax+by+d=0 \quad \text{respectively.}$$

Imp. Hence we can say that the plane $ax+by+cz+d=0$ represents planes perpendicular to yz , zx , xy planes respectively according as $a=0$ or $b=0$ or $c=0$ or according as the term of x , or of y or of z is missing in its equation.

5. Planes parallel to the co-ordinate axes.

Any plane perpendicular to the yz -plane is parallel to the axis of x and hence equation to the planes parallel to the co-ordinate axes is the same as the equation of the planes perpendicular the co-ordinate planes (Case 4).

Ex. 9. - Determine the three planes through the intersection of the planes $x+y+z=1$ and $2x+3y-z+4=0$ which are parallel to the three co-ordinate axes. Also find the equation of the plane perpendicular to the yz -plane and passing through the points $(2, 3, 1)$ and $(4, -5, 3)$.

Any plane through the intersection of the given planes is

$$(x+y+z-1)+\lambda(2x+3y-z+4)=0$$

$$\text{or} \quad x(1+2\lambda)+y(1+3\lambda)+z(1-\lambda)-(1-4\lambda)=0. \quad \dots (1)$$

If it is parallel to x -axis, i.e. perpendicular to yz -plane, then by case 4 the coefficient of x should be zero.

$$\text{i.e.} \quad 1+2\lambda=0 \quad \text{or} \quad \lambda=-\frac{1}{2}.$$

Putting the value of λ in (1) we get the equation of the required plane as $y-3z+6=0$.

Similarly the equation of planes parallel to the other two axes or perpendicular to other two co-ordinate planes are $x+4z=7$ and $3x+4y+3=0$.

Again general equation of the plane perpendicular to yz plane or parallel to the axis of x is $by+cz+d=0$ $\therefore (a=0)$.

Since it passes through the given points, we have

$$3b + c + d = 0 \text{ and } -5b + 3c + d = 0$$

$$\therefore \frac{b}{1-3} = \frac{c}{-5-3} = \frac{d}{9+5} \text{ or } \frac{b}{1-3} = \frac{c}{-8} = \frac{d}{14}$$

$\therefore$ the required equation of the plane is $y + 4z - 7 = 0$.

(b) Find the equation of the plane through the line of intersection of the planes $P = ax + by + cz + d = 0$ and $Q = a'x + b'y + c'z + d' = 0$, under the following conditions

(i) perpendicular to xy -plane i.e. $z = 0$ [Kurkshetra 73]

(ii) Parallel to x -axis i.e. perpendicular to yz -plane.

(iii) perpendicular to plane $lx + my + nz + p = 0$ [Indore 79]

The required plane is $P + \lambda Q = 0$

$$\text{or } (ax + by + cz + d) + \lambda (a'x + b'y + c'z + d') = 0 \quad \dots(1)$$

$$\text{or } x(a + \lambda a') + y(b + \lambda b') + z(c + \lambda c') + (d + \lambda d') = 0 \quad \dots(1)$$

(i) It is perpendicular to $z = 0$ $\therefore$ coefficient of $z = 0$

i.e. It will not have the term of z

$$\therefore c + \lambda c' = 0 \text{ or } \lambda = -c/c'$$

Hence from (2) the plane is

$$(ac' - a'c)x + (bc' - b'c)y + (dc' - d'c) = 0$$

$$(ii) (ab' - a'b)y + (ac' - a'c)z + (ad' - a'd) = 0$$

$$(iii) \text{ This is perpendicular to plane } lx + my + nz = p$$

Applying the condition of perpendicularity of the two planes we get

$$l(a + \lambda a') + m(b + \lambda b') + n(c + \lambda c') = 0$$

$$\therefore \lambda = -\frac{(al + bm + cn)}{(a'l + b'm + c'n)}$$

Putting in (1) the equation of the plane is

$$P(a'l + b'm + c'n) = Q(al + bm + cn).$$

§ 10. Position of a point with respect to a plane.

To find the ratio in which the line joining two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is divided by the plane $Ax + By + Cz + D = 0$ and find the condition for the given points to lie on the same side or opposite sides of the given plane.

Let the plane divide the join of two points in the ratio $\lambda : 1$ at the point P so that its co-ordinate are

$$\left(\frac{\lambda x_2 + x_1}{\lambda + 1}, \frac{\lambda y_2 + y_1}{\lambda + 1}, \frac{\lambda z_2 + z_1}{\lambda + 1} \right) \quad [\text{Cor 2. P. 11}]$$

Since the point P lies on the plane $Ax + By + Cz + D = 0$,

$$\text{we have } A \frac{\lambda x_2 + x_1}{\lambda + 1} + B \frac{\lambda y_2 + y_1}{\lambda + 1} + C \frac{\lambda z_2 + z_1}{\lambda + 1} + D = 0$$

$$\text{or } \lambda (Ax_2 + By_2 + Cz_2 + D) + (Ax_1 + By_1 + Cz_1 + D) = 0,$$

$$\text{or } \lambda = -\frac{Ax_1 + By_1 + Cz_1 + D}{Ax_2 + By_2 + Cz_2 + D}.$$

Now if λ is +ve, then the plane divides the line joining the given points internally, i.e., the two points lie on the opposite sides of the plane. But if λ is -ve then from (1) we conclude that $Ax_1 + By_1 + Cz_1 + D$ and $Ax_2 + By_2 + Cz_2 + D$ should be of opposite signs.

Again if λ is -ve, then the plane divides the line joining the given points externally, i.e., the two points lie on the same side of the plane. Now if λ is +ve, then from (1) we conclude that $Ax_1 + By_1 + Cz_1 + D$ and $Ax_2 + By_2 + Cz_2 + D$ are of the same sign.

Hence we conclude that two points (x_1, y_1, z_1) and (x_2, y_2, z_2) lie on the same side or opposite sides of the plane according as $Ax_1 + By_1 + Cz_1 + D$ and $Ax_2 + By_2 + Cz_2 + D$ are of same or of opposite signs.

Cor. In the equation of the plane $Ax + By + Cz + D = 0$ if we take the constant term D positive, then the origin and the point (x_1, y_1, z_1) lie on the same side as the origin if $Ax_1 + By_1 + Cz_1 + D$ is +ive. In case it be -ive, then the point (x_1, y_1, z_1) and the origin lie on the opposite sides.

Note. In case D be not +ive, then we can make it +ive, by changing the sign throughout so that when $(0, 0, 0)$ is put in the equation of the plane, the result is +ive and equal to D .

Ex. 10. Find whether the two points $(2, 3, -5)$ and $(3, 4, 7)$ lie on the same side or opposite sides of the plane

$$x + 2y - 2z - 9 = 0$$

Substituting the co-ordinates of the given points in the equation of the plane we get +9 and -12. Since the results are of opposite signs, the two points lie on the opposite sides of the given plane.

§ 11. Perpendicular distance of a point from a given plane.

[Sagar 65; Cal. 81; Raj. 78; Ranchi 78; Agra 66; Madras 74; Marthwada 74; Mysore; 71]

Case 1. Normal form.

Let the equation of the plane in normal form be

$$lx + my + nz = p. \quad \dots(1)$$

where p stands for the length of the perpendicular from the origin i.e. ON and let the given point be (x_1, y_1, z_1) .

Equation of a plane parallel to (1) is

$$lx + my + nz = p_1, \quad \dots(2)$$

and since it passes through (x_1, y_1, z_1) , we have

$$p_1 = lx_1 + my_1 + nz_1.$$

$p_1 = ON'$ is the length of the perpendicular from origin to the plane (2) i.e. the plane through P and parallel to given plane.

Now if PL be drawn perpendicular to the given plane from the point P , then

$$LP = ON' - ON = p_1 - p = (lx_1 + my_1 + nz_1 - p)$$

Working Rule. Substitute the co-ordinates of the given point in the given equation of the plane. (Normal form)

Case II. The equation of the plane being in general form.

Let the equation of the plane in general form be

$$Ax + By + Cz + D = 0.$$

By § 5 the corresponding normal form of the above according as D is -ive or +ive is

$$\begin{aligned} & \pm \frac{A}{\sqrt{A^2 + B^2 + C^2}} x \pm \frac{B}{\sqrt{A^2 + B^2 + C^2}} y \\ & \pm \frac{C}{\sqrt{A^2 + B^2 + C^2}} z = \pm \frac{D}{\sqrt{A^2 + B^2 + C^2}} \\ & \pm \frac{A}{\sqrt{A^2 + B^2 + C^2}} x \pm \frac{B}{\sqrt{A^2 + B^2 + C^2}} y \\ & \pm \frac{C}{\sqrt{A^2 + B^2 + C^2}} z \pm \frac{D}{\sqrt{A^2 + B^2 + C^2}} = 0 \end{aligned}$$

Hence if d be the perpendicular distance of (x_1, y_1, z_1) then by

$$\text{Case I, } d = \pm \frac{Ax_1 + By_1 + Cz_1 + D}{\sqrt{A^2 + B^2 + C^2}}.$$

In other words, substitute the co-ordinates of the given point and divide by

$$\sqrt{\{(\text{coeff. of } x)^2 + (\text{coeff. of } y)^2 + (\text{coeff. of } z)^2\}}$$

Note. The above rule is identical with the corresponding rule of plane geometry.

Cor. Distance between two parallel planes.

First Method

Choose any point on one of the planes and find its perpendicular distance from the other.

Second Method.

Find out the perpendicular distance of each plane from the origin and retain their signs and subtract the two.

Ex. 11. Find the distance of the points $(2, 0, 1)$ and $(3, -3, 2)$ from the plane $x - 2y + z = 6$, and find whether the two points lie on the same side or opposite sides of the plane.

The equation of the plane is $x - 2y + z - 6 = 0$ (1)

Its distance from $(2, 0, 1)$ is $\frac{2 - 2 \cdot 0 + 1 - 6}{\sqrt{(1^2 + 2^2 + 1^2)}} = \frac{-3}{\sqrt{6}} = -\frac{3}{\sqrt{6}}$.

Its distance from $(3, -3, 2)$ is $\frac{3 - 2(-3) + 2 - 6}{\sqrt{(1^2 + 2^2 + 1^2)}} = \frac{5}{\sqrt{6}}$.

Again the two points when substituted in the L.H.S. of (1) give the results of opposite signs. Therefore the two points lie on the opposite sides. [§ 10 p. 48]

Exercise 4

1. Find the intercepts made by the plane $2x + y - 2z = 3$ on the co-ordinate axes. What are the d.c's of the normal to the plane?

The D. R's of the normal to the plane are given by coefficients of x, y, z in the equation of the plane which are $2, 1, -2$ and hence d.c's are $\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}$. Also the given equation can be put as

$$\frac{x}{\frac{3}{2}} + \frac{y}{3} + \frac{z}{-\frac{3}{2}} = 1. \quad \text{Intercepts are } \frac{3}{2}, 3, -\frac{3}{2}.$$

2. If a variable plane moves in such a way that the sum of the reciprocals of its intercepts on the axes is constant, then prove that the plane passes through a fixed point.

Let the equation of the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$... (1)

whose intercepts are a, b, c . It is given that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \text{constant} = \frac{1}{k} \text{ say.}$$

$$\therefore \frac{k}{a} + \frac{k}{b} + \frac{k}{c} = 1$$

This relation shows that the plane (1) passes through the point (k, k, k) which is fixed.

3. Prove that the equation of the plane through the points $(1, 1, 1)$, $(1, -1, 1)$ and $(-7, -3, -5)$ is parallel to the axis of y , i.e. perpendicular to xz -plane.

Any plane through $(1, 1, 1)$ is

$$A(x-1) + B(y-1) + C(z-1) = 0.$$

If it passes through the other two given points then

$$0A - 2B + 0C = 0 \quad \therefore B = 0;$$

and

$$-8A - 6C = 0$$

$$\therefore \frac{A}{3} = \frac{C}{-4} = k \text{ and } B=0.$$

Hence from (1) the required equation of the plane is

$$3(x-1) - 4(z-1) = 0 \text{ or } 3x - 4z + 1 = 0. \quad \dots(2)$$

Since the term of y is missing and as such the plane (2) is perpendicular to $y=0$, i.e. xz plane. Otherwise also the plane $y=0$, has D. R's of normal as 0, 1, 0 and the plane (2) has D. R's of normal as 3, 0, -4 and since $3 \cdot 0 + 1 \cdot 0 - 4 \cdot 0 = 0$ i.e. $\Sigma a_1 a_2 = 0$ and hence perpendicular.

4. - Determine the constant k , so that the planes $x - 2y + kz = 0$ and $2x + 5y - z = 0$ are at right angles, and find in that case the plane through the point $(1, -1, -1)$ and perpendicular to both the given planes.

If the planes are perpendicular, then

$$A_1 A_2 + B_1 B_2 + C_1 C_2 = 0$$

$$\therefore 1 \cdot 2 - 2 \cdot 5 + k(-1) = 0 \quad \therefore k = -8.$$

Any plane through $(1, -1, -1)$ is

$$A(x-1) + B(y+1) + C(z+1) = 0$$

Since it is perpendicular, to both the given planes ($k = -8$), we have

$$A \cdot 1 + B(-2) + C(k) = 0 \text{ or } A - 2B - 8C = 0$$

$$A \cdot 2 + B \cdot 5 + C(-1) = 0 \text{ or } 2A + 5B - C = 0$$

$$\therefore \frac{A}{2+40} = \frac{B}{-16+1} = \frac{C}{9} \text{ or } \frac{A}{14} = \frac{B}{-5} = \frac{C}{3} = k \text{ say,}$$

$$\text{Hence the plane is } 14(x-1) - 5(y+1) + 3(z+1) = 0$$

$$\text{or } 14x - 5y + 3z = 16.$$

(b) Find the equation of the plane through $(-1, 3, 2)$ and perpendicular to each of the planes.

$$x + 2y + 3z = 5 \text{ and } 3x + 3y + z = 9$$

$$\text{Ans. } 7x - 8y + 3z + 25 = 0 \quad [\text{Andhra 84; Jodhpur 78; Meerut 80; Venka 84; Kanpur 80}]$$

(5) Find the equation of the plane through the points $(-1, 1, 1)$ and $(1, -1, 1)$ and perpendicular to the plane $x + 2y + 2z = 7$.

[Bhagalpur 75; Bihar 73]

Any plane through $(-1, 1, 1)$ is

$$A(x+1) + B(y-1) + C(z-1) = 0 \quad \dots(1)$$

It passes through $(1, -1, 1)$

$$\therefore 2A - 2B + 0C = 0. \quad \dots(2)$$

It is perpendicular to the plane

*

$$x + 2y + 2z = 7$$

$$A + 2B + 2C = 0.$$

Solving (2) and (3), we get

$$\frac{A}{-4} = \frac{B}{-4} = \frac{C}{6} \text{ or } \frac{A}{2} = \frac{B}{2} = \frac{C}{-3} = k, \text{ say}$$

and hence A, B, C are proportional to $2, 2, -3$ and putting in (1) the equation of the plane is

$$2(x+1) + 2(y-1) - 3(z-1) = 0$$

$$2x + 2y - 3z + 3 = 0$$

or

(b) Find the equation of the plane through the points $(2, 2, 1)$ and $(9, 3, 6)$ and perpendicular to the plane

$$2x + 6y + 6z = 9$$

[Rohailkhand 82; Gorakhpur 80; Meerut 78; Madras 78; Agra 77; Avadh 79]

Proceed exactly as in part (a)

$$\frac{A}{3} = \frac{B}{4} = \frac{C}{-5} = k, \quad 3x + 4y - 5z = 9.$$

6. Prove that the equation of the plane through the points $(1, -2, 4)$ and $(3, -4, 5)$ and perpendicular to yz -plane or parallel to x -axis is $y + 2z = 6$. [Agra 82; Gorakhpur 76; Raj. 84]

Any plane perpendicular to yz -plane, i.e. $x=0$ shall not contain the term of x in it (P. 47). Hence let its equation be

$$by + cz + d = 0 \quad \dots (1)$$

It passes through $(1, -2, 4)$ and $(3, -4, 5)$;

$$-2b + 4c + d = 0$$

$$-4b + 5c + d = 0$$

or

$$\frac{b}{-1} = \frac{c}{-2} = \frac{d}{6}$$

Putting in (1), we get

$$-y - 2z + 6 = 0 \text{ or } y + 2z = 6$$

7. Find the equation of the plane which bisects the line joining the points $(1, 2, 3)$ and $(3, 4, 5)$ at right angles and show that it makes equal intercepts on the axes. [Burdwan 79]

D.R's of the given line are $3-1, 4-2, 5-3$ or $1, 1, 1$. Since the line is at right angles to the plane and as such it is parallel to the normal of the plane. Also it passes through mid. point $(2, 3, 4)$ of the given line. Hence its equation is

$$1(x-2) + 1(y-3) + 1(z-4) = 0 \text{ or } x + y + z = 9$$

8. (a) Find the equation of the plane that passes through (2, -3, 1) and is normal to the line joining the points (3, 4, -1) and (2, -1, 5). (Ans. $x+5y-6z+19=0$)

[Agra 78, Meerut 79, Magadh 74]

(b) Prove that $3x-y-z+11=0$ is the equation of the plane through (-1, 6, 2) and perpendicular to the join of the points (1, 2, 3) and (-2, 3, 4). [Madras 77]

9. Find the equation of the plane through the point (1, 2, 3) and parallel to the plane $3x+2y+z=7$. (Ans. $3x+2y+z=10$)

10. Find the equation of the plane through the intersection of the planes $x+2y+3z+4=0$ and $4x+3y+3z+1=0$ and perpendicular to the plane $x+y+z+9=0$, and show that it is perpendicular to xy -plane. [See Ex. 8 P. 45]

Ans. $x-z=2$. The coefficient of y is zero; hence it is perpendicular to zx -plane. [§ 9 Case 4 P. 47]

11. Find the equation of the plane through the intersection of the planes $x-2y+3z+4=0$ and $2x-3y+4z-7=0$ and the point (1, -1, 1). (Ans. $9x-13y+17z-39=0$)

Use $P+\lambda Q=0$ and find λ by passing through (1, -1, 1).

12. Find the equation of the plane through the intersection of the planes $x+3y+6=0$ and $3x-y-4z=0$, whose perpendicular distance from the origin is unity. [Rohailkhand 77]

Any plane through the intersection of given planes is

$$(x+3y+6)+\lambda(3x-y-4z)=0$$

$$\text{or } (1+3\lambda)x+(3-\lambda)y-4\lambda z+6=0. \quad \dots(1)$$

Its perpendicular distance from (0, 0, 0) is (1)

$$\therefore \frac{6}{[(1+3\lambda)^2+(3-\lambda)^2+16\lambda^2]^{1/2}}=1.$$

$$\text{or } 1+6\lambda+9\lambda^2+9+\lambda^2-6\lambda+16\lambda^2=36$$

$$\therefore 26\lambda^2=26 \text{ or } \lambda=\pm 1.$$

Putting the values of λ in (1), the required equations are

$$2x+y-2z+3=0 \text{ and } x-2y-2z-3=0.$$

13. Prove that the points (1, -1, 3) and (3, 3, 3) are equidistant from the plane $5x+2y-7z+9=0$ and on opposite sides of it.

14. Find the distances of the points (1, -2, 3) and (2, 3, 3) from the plane $x-2y+2z=5$. Are the points on the same side of the plane?

Ans. $p_1=2, p_2=1$. The points are on the opposite sides of the given plane

15. Are the two planes given by the equations $3x+4y+5z+10=0$ and $9x+12y+15z+20=0$ parallel? If so find the distance between them.

Since $\frac{3}{9} = \frac{4}{12} = \frac{5}{15}$ i.e. the coefficients of x, y, z in the two equations are proportional and as such the planes are parallel. The distance between them is $p_1 - p_2$ where p_1 and p_2 are their distances from origin with sign retained.

$$p_1 = \frac{10}{\sqrt{9+16+25}} = \frac{10}{5\sqrt{2}} = \sqrt{2}.$$

$$p_2 = \frac{20}{3\sqrt{9+16+25}} = \frac{20}{15\sqrt{2}} = \frac{2\sqrt{2}}{3}.$$

$$\therefore p_1 - p_2 = \sqrt{2} - \frac{2\sqrt{2}}{3} = \frac{\sqrt{2}}{3}.$$

(b) Prove that the distance between parallel planes $2x - 2y + z + 3 = 0$ and $4x - 4y + 2z + 5 = 0$ is $1/6$. [Berhampur 81]

Some Typical Solved Examples

Ex. 1. A variable plane passes through a fixed point (a, b, c) and meets the axes of reference in A, B and C . Show that the locus of the point of intersection of the planes through A, B and C parallel to the co-ordinate planes is [Utkal 82; Madras 74]

$$ax^{-1} + by^{-1} + cz^{-1} = 1.$$

Let the plane be $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$.

It passes through (a, b, c) ; $\therefore \frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} = 1$ (1)

It meets the axes in $A(\alpha, 0, 0), B(0, \beta, 0), C(0, 0, \gamma)$.

Also planes through A, B and C and parallel respectively to the co-ordinate planes are

$x = \alpha, y = \beta, z = \gamma$ respectively. ... (2)

The locus of the point of intersection of the planes is obtained by eliminating the variables α, β, γ .

Putting for α, β, γ from (2) in (1), we get the required locus as $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$ or $ax^{-1} + by^{-1} + cz^{-1} = 1$.

Ex. 2. A variable plane at a constant distance p from the origin meets the axes in A, B and C . Through A, B, C planes are drawn parallel to the co-ordinate planes; show that locus of their point of intersection is

$$x^{-2} + y^{-2} + z^{-2} = p^{-2}.$$

[Vikram 84; Garhwal 83, 75; Indore 70; Rohailkhand 81; Kanpur 75; Rajputana 83; Meerut 74]

Let the equation of the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

It meets the axes in $A(a, 0, 0)$, $B(0, b, 0)$, $C(0, 0, c)$. Its distance from origin is p given.

$$\therefore \frac{1}{\sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right)}} = p \text{ or } \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad \dots (1)$$

Planes through A, B, C and parallel to co-ordinate planes are $x=a$, $y=b$ and $z=c$ respectively. $\dots (2)$

Locus of their point of intersection is obtained by eliminating the variables a, b, c between (1) and (2).

Putting for a, b and c from (2) in (1), we get the required locus as $\frac{1}{p^2} = \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}$ or $x^{-2} + y^{-2} + z^{-2} = p^{-2}$.

Ex. 3 A variable plane is at a constant distance p from the origin and meets the axes in A, B and C . Show that the locus of the centroid of the tetrahedron $OABC$ is

$$x^{-2} + y^{-2} + z^{-2} = 16p^{-2}.$$

[Lucknow 82; Ranchi 74; Raj. 81, 74; Kerala 69; Agra 82, 77; Kanpur 79; Meerut 83]

Let the plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

Its distance from origin is p given.

$$\frac{1}{\sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right)}} = p \text{ or } \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad \dots (1)$$

Also the points O, A, B, C are respectively.

$(0, 0, 0)$, $(a, 0, 0)$, $(0, b, 0)$ and $(0, 0, c)$.

$$x = \frac{0+a+0+0}{4} \text{ or } x = \frac{a}{4}, y = \frac{b}{4}, z = \frac{c}{4}.$$

$$\therefore a = 4x, b = 4y, c = 4z.$$

Putting for a, b, c in (1), we get the locus of centroid as

$$\frac{1}{p^2} = \frac{1}{16x^2} + \frac{1}{16y^2} + \frac{1}{16z^2} \text{ or } x^{-2} + y^{-2} + z^{-2} = 16p^{-2}.$$

Ex. 4 A variable plane is at a constant distance $3p$ from the origin and meets the axes in A, B and C . Show that the locus of the centroid of the triangle ABC is

$$x^{-2} + y^{-2} + z^{-2} = p^{-2}.$$

[Meerut 83; Agra 78; Indore 84; Madras 75; Punjab 76; Madurai 82; Luck. 80; Utkal 81; Rohailkhand 79]

As above we have

$$\frac{1}{\sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right)}} = 3p \therefore \frac{1}{9p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad \dots (1)$$

If (x, y, z) be the centroid of $\triangle ABC$, then

$$x = \frac{a+0+0}{3} = \frac{a}{3}, y = \frac{b}{3}, z = \frac{c}{3};$$

$$\therefore a = 3x, b = 3y \text{ and } c = 3z.$$

Putting for a, b, c in (1) we get the locus as given.

Ex. 5. A point P moves on the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ which is fixed. The plane through P perpendicular to OP meets the axes in A, B and C . The planes through A, B, C parallel to yz, zx, xy planes intersect in Q . Prove that if the axes be rectangular, the locus of Q is

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}.$$

[Kanpur 83; Raj. 77; Jiwaji 66; Agra 65]

If the co-ordinates of the point P be (α, β, γ) , then

$$\frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = 1. \quad \dots (1)$$

Again direction cosines of OP are proportional to α, β, γ and hence these are also the direction-ratios of the normal to the plane which is perpendicular to OP and since it passes through P , its equation is

$$\alpha(x - \alpha) + \beta(y - \beta) + \gamma(z - \gamma) = 0 \quad (\text{See Ex. 5 p. 44})$$

$$\text{or} \quad \alpha x + \beta y + \gamma z = \alpha^2 + \beta^2 + \gamma^2. \quad \dots (2)$$

It meets the axes in A, B, C and hence the co-ordinates of these points are $\left(\frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}, 0, 0\right)$ etc. The equation of the

plane through A and parallel to the yz -plane is $x = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha}$.

Similarly the equations of other planes are $y = \frac{\alpha^2 + \beta^2 + \gamma^2}{\beta}$ and

$$z = \frac{\alpha^2 + \beta^2 + \gamma^2}{\gamma}.$$

The locus of their point of intersection is obtained by eliminating α, β, γ between the three equations of the planes and relation (1).

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{\alpha^2 + \beta^2 + \gamma^2}{(\alpha^2 + \beta^2 + \gamma^2)^2} = \frac{1}{\alpha^2 + \beta^2 + \gamma^2}$$

$$\text{Again } \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz} = \frac{\alpha/a + \beta/b + \gamma/c}{(\alpha^2 + \beta^2 + \gamma^2)^2} = \frac{1}{\alpha^2 + \beta^2 + \gamma^2} \text{ from (1)}$$

$$\therefore \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{ax} + \frac{1}{by} + \frac{1}{cz} \text{ is the required locus.}$$

Ex. 6. Two systems of rectangular axes have the same origin. If a plane cuts them at distances a, b and c and a', b', c' respectively from the origin, prove that

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$$

[Rohailkhand 81; L.N.M. 82; 81, 74; Kanpur 82, 79]

Let the equation of the plane with respect to the first system be $x/a + y/b + z/c = 1$, the axes being OX, OY and OZ .

Similarly the equation of the same plane with respect to the second system is $\xi/a' + \eta/b' + \zeta/c' = 1$; the axes being $(O\xi, O\eta, O\zeta)$.

Now from a point outside a plane only one perpendicular can be drawn and hence the length of the perpendicular from origin to the plane written w.r.t. two different systems of rectangular axes must be the same.

$$\therefore \frac{1}{\sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right)}} = \frac{1}{\sqrt{\left(\frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}\right)}}$$

or

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2}$$

Ex. 7. Find the equation of two planes through the points $(0, 4, -3)$ and $(6, -4, 3)$ other than the planes through the origin, which cut off from the axes intercepts whose sum is zero.

Let the equation of the required plane be

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1, \text{ where } a + b + c = 0.$$

...(1)

Now substituting the given points, we get

$$\frac{4}{b} - \frac{3}{c} = 1 \quad \dots(2) \quad \text{and} \quad \frac{6}{a} - \frac{4}{b} + \frac{3}{c} = 1.$$

...(2)

Adding (2) and (3) we get $\frac{6}{a} = 2$.

$$\therefore a = 3 \text{ and subtracting we get } 2\left(\frac{4}{b} - \frac{3}{c}\right) = 2.$$

*

or $4c-3b=bc$. Also $b+c=-a=-3$; $\therefore c=-3-b$
 $\therefore 4(-3-b)-3b=b(-3-b)$ or $-12-7b=-3b-b^2$
 or $b^2-4b-12=0$ or $(b-6)(b+2)=0$; $\therefore b=6, -2$
 and $c=-9, -1$,
 i.e. $a=3, b=6, c=-9$ or $a=3, b=-2, c=-1$, etc.

Ans. $6x+3y-2z=18, 2x-3y-6z=6$.

Ex. 8. From any point P are drawn PM and PN perpendiculars to zx and xy -planes. O is the origin and α, β and θ are the angles which OP makes with the co-ordinate planes and with the plane OMN . Prove that if the co-ordinates of the point P are (a, b, c) , then

(a) The plane OMN is $\frac{x}{a} - \frac{y}{b} - \frac{z}{c} = 0$.

(b) $\theta = \sin^{-1} \frac{abc}{\sqrt{(a^2+b^2+c^2)} \sqrt{(b^2c^2+c^2a^2+a^2b^2)}}$

(c) $\operatorname{cosec}^2 \theta = \operatorname{cosec}^2 \alpha + \operatorname{cosec}^2 \beta + \operatorname{cosec}^2 \gamma$.

M is the foot of the perpendicular from (a, b, c) on the zx -plane and hence this point is $(a, 0, c)$. Similarly N in the xy -plane is $(a, b, 0)$ and $O(0, 0, 0)$.

Any plane through $(0, 0, 0)$ is $Ax + By + Cz = 0$... (1)

It passes through M and N ; we have

$Aa + Bb + C \cdot 0 = 0$ and $Aa + 0 \cdot b + Cc = 0$.

$\therefore \frac{A}{bc} = \frac{B}{-ac} = \frac{C}{-ab}$

Hence from (1) the equation of the plane OMN is

$bcx - acy - abz = 0$ or $\frac{x}{a} - \frac{y}{b} - \frac{z}{c} = 0$.

Again direction cosines of OP are proportional to $a-0, b-0, c-0$, i.e. a, b, c and the direction cosines of the normal to plane OMN are proportional to $\frac{1}{a}, -\frac{1}{b}, -\frac{1}{c}$. Now if θ is the angle between the line OP and the plane OMN , then $90-\theta$ is the angle between the line OP and the normal to the plane OMN whose direction-ratios are written above.

$\therefore \cos(90-\theta) = \frac{a\left(\frac{1}{a}\right) + b\left(-\frac{1}{b}\right) + c\left(-\frac{1}{c}\right)}{\sqrt{(a^2+b^2+c^2)} \sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right)}} \quad [\text{P. 24}]$

or $\sin \theta = \frac{-abc}{\sqrt{(a^2+b^2+c^2)} \sqrt{(b^2c^2+c^2a^2+a^2b^2)}}$

Neglecting -ive sign we have

$$\theta = \sin^{-1} \frac{abc}{\sqrt{(a^2+b^2+c^2)}\sqrt{(b^2c^2+c^2a^2+a^2b^2)}} \quad \dots(2)$$

Again α is the angle between the line OP and the yz -plane i.e. $x=0$ and hence $90-\alpha$ is the angle between line OP and the normal to the plane $x=0$ whose direction ratios are $(1, 0, 0)$.

$$\therefore \cos(90-\alpha) = \frac{a \cdot 1 + b \cdot 0 + c \cdot 0}{\sqrt{(a^2+b^2+c^2)}\sqrt{(1^2+0^2+0^2)}}$$

$$\text{or } \sin \alpha = \frac{a}{\sqrt{(a^2+b^2+c^2)}} \text{ or } \operatorname{cosec}^2 \alpha = \frac{a^2+b^2+c^2}{a^2}.$$

$$\text{Similarly, } \operatorname{cosec}^2 \beta = \frac{a^2+b^2+c^2}{b^2} \text{ and } \operatorname{cosec}^2 \gamma = \frac{a^2+b^2+c^2}{c^2}$$

$$\begin{aligned} \therefore \operatorname{cosec}^2 \alpha + \operatorname{cosec}^2 \beta + \operatorname{cosec}^2 \gamma &= (a^2+b^2+c^2) \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) \\ &= \frac{(a^2+b^2+c^2)(b^2c^2+c^2a^2+a^2b^2)}{a^2b^2c^2} = \operatorname{cosec}^2 \theta \quad \text{from (2)} \end{aligned}$$

Ex. 9. Find the equation of the plane which bisects the join of $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ perpendicularly.

Direction cosines of the line PQ are proportional to $x_1-x_2, y_1-y_2, z_1-z_2$. Since the required plane is perpendicular to the line, hence its normal is parallel to the line PQ : therefore the direction ratios of the normal to the plane are

$$x_1-x_2, y_1-y_2, z_1-z_2.$$

Hence equation of the plane is

$$(x_1-x_2)x + (y_1-y_2)y + (z_1-z_2)z = D. \quad \dots(1)$$

Since the plane bisects PQ , it passes through the mid-point of PQ i.e. $\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}$. Putting in (1), we get

$$D = \frac{1}{2} \{ (x_1^2 - x_2^2) + (y_1^2 - y_2^2) + (z_1^2 - z_2^2) \}.$$

Hence the equation of the plane is

$$\Sigma x(x_1-x_2) = \frac{1}{2} \Sigma (x_1^2 - x_2^2).$$

Ex. 10. A plane meets a set of three mutually perpendicular planes in the sides of a triangle whose angles are A, B and C respectively. Show that the first plane makes with the other planes angles, the square of whose cosines are

$$\cot B \cot C, \cot C \cot A, \cot A \cot B.$$

Let the three mutually perpendicular planes be co-ordinate planes and any plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

... (1)

meets at $A(a, 0, 0)$, $B(0, b, 0)$ and $C(0, 0, c)$.

D.R.'s of AB are $a, -b, 0$ and D.R.'s of AC are $a, 0, -c$.

Angle between sides AB and AC is A .

$$\therefore \tan A = \frac{[\Sigma (b_1 c_2 - b_2 c_1)^2]^{1/2}}{a_1 a_2 + b_1 b_2 + c_1 c_2} \text{ in terms of direction ratios}$$

$$\tan A = \frac{\sqrt{(b^2 c^2 + c^2 a^2 + a^2 b^2)}}{a^2}$$

$$\therefore \cot A = \frac{a^2}{\sqrt{(a^2 b^2 + b^2 c^2 + c^2 a^2)}}$$

$$\text{Similarly } \cot B = \frac{b^2}{(a^2 b^2 + b^2 c^2 + c^2 a^2)^{1/2}}$$

$$\text{and } \cot C = \frac{c^2}{(a^2 b^2 + b^2 c^2 + c^2 a^2)^{1/2}}$$

Let α be the angle between plane (1) and one of the co-ordinate planes say $x=0$. The D.R.'s of normals to these planes are $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ and $1, 0, 0$ respectively.

$$\therefore \cos \alpha = \frac{\frac{1}{a} \cdot 1 + 0 + 0}{\sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}\right)}} \sqrt{1}$$

$$\text{or } \cos^2 \alpha = \frac{b^2 c^2}{a^2 b^2 + b^2 c^2 + c^2 a^2} = \cot B \cot C.$$

Similarly $\cos^2 \beta = \cot C \cot A$ and $\cos^2 \gamma = \cot A \cot B$.

Ex. 11. A triangle the length of whose sides are a, b and c is placed so that the middle points of the sides are on the axes. Prove that the equation to its plane is

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$$

where $8\alpha^2 = b^2 + c^2 - a^2$, $8\beta^2 = c^2 + a^2 - b^2$ and $8\gamma^2 = a^2 + b^2 - c^2$ and

that the co-ordinates of the vertices are

$$(-\alpha, \beta, \gamma), (\alpha, -\beta, \gamma), (\alpha, \beta, -\gamma) \quad (\text{Meerut 70; Agra 38})$$

If α, β, γ be the intercepts on the axes, then the equation of

$$\text{the plane is } \frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1.$$

The points on the axes are $(\alpha, 0, 0)$, $(0, \beta, 0)$, $(0, 0, \gamma)$

Now we know that the line joining the mid. points of the sides of a triangle is parallel to the third side and half its length

$$\sqrt{(\alpha^2 + \beta^2)} = \frac{c}{2} \text{ or } \alpha^2 + \beta^2 = \frac{c^2}{4}.$$

Similarly $\beta^2 + \gamma^2 = \frac{a^2}{4}$ and $\gamma^2 + \alpha^2 = \frac{b^2}{4}$.

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = \frac{a^2 + b^2 + c^2}{8}$$

$$\therefore \alpha^2 = \frac{a^2 + b^2 + c^2}{8} - \frac{a^2}{4} = \frac{b^2 + c^2 - a^2}{8} \text{ etc.}$$

Since mid. point of BC is $(\alpha, 0, 0)$.

$\therefore$ x-co-ordinate

of both B and C should be α, α .

Similarly mid point of CA is $(0, \beta, 0)$;

$\therefore$ y-co-ordinate

of both C and A should be β, β .

Similarly z-co-ordinate of both A and B should be γ, γ .

The remaining co-ordinate is found because the other co-ordinate of mid. points are all zero.

§ 12. Equations of the planes bisecting the angle between the given planes. [Garhwal 81, 79; Meerut 71; Ravi Shanker 70]

Let the equations of the two given planes be

$$a_1x + b_1y + c_1z + d_1 = 0 \quad \dots(1)$$

and

$$a_2x + b_2y + c_2z + d_2 = 0$$

If (x, y, z) be the co-ordinates of any point on the plane bisecting the angle between the given planes, then by definition the perpendicular distances of the point from the two planes are equal numerically.

$$\therefore \frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = \pm \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Simplifying we get the equations of the required planes bisecting the angle between the given planes.

How to distinguish between the two planes.

(i) If we are to determine which of the two planes represents the plane bisecting the acute or obtuse angle between the planes, we should find the value of $\cos \theta$ where θ is the angle between the bisecting plane and any of the given planes. From the value of $\cos \theta$ obtained find the value of $\tan \theta$. In case the value of $\tan \theta$ is greater than 1, then $\theta > 45^\circ$ and hence that plane bisects the obtuse angle between them. In case the value of $\tan \theta$ is less than 1, then $\theta < 45^\circ$ and hence that plane bisects the acute angle between the planes.

(ii) Sometimes we required to find which of the planes bisects the angle between the given planes that contains the origin.

For that we should express the equations of the planes in the form so that the constant terms in both of them are +ive ; then the bisecting plane

$$\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = + \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

bisects the angle between the planes that contains the origin and the plane

$$\frac{a_1x + b_1y + c_1z + d_1}{\sqrt{a_1^2 + b_1^2 + c_1^2}} = - \frac{a_2x + b_2y + c_2z + d_2}{\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

bisects the angle between the planes that does not contain the origin.

Combined Method.

If the origin lies in the acute angle between the planes, then the angle between the normals to the planes is obtuse and hence the value of $\cos \theta$ is -ive.

If $a_1a_2 + b_1b_2 + c_1c_2 = -\text{ive}$ then origin lies in the acute angle. Similarly if origin lies in the obtuse angle between the planes then the angle between the normals to the planes is acute hence the value of $\cos \theta$ is +ive.

$\therefore$ If $a_1a_2 + b_1b_2 + c_1c_2 = +\text{ive}$, then the origin lies in the obtuse angle.

Note—The equations of the planes should be written in the form so that the constant terms in both of them are of the same sign before calculating the value of $a_1a_2 + b_1b_2 + c_1c_2$. Hence while distinguishing between the two bisecting planes we should first find whether the origin lies in the acute angle or obtuse angle by the above method ; then we should find the bisecting plane which contains the origin by method No. II. In this way we shall be able to distinguish between the two bisecting planes with respect to the position of the origin and the acute and obtuse angles.

Exercise 5

Ex. 1. Find the equations of the planes bisecting the angle between the planes $x + 2y + 2z = 9$ and $4x - 3y + 12z + 13 = 0$ and distinguish them.

[Avadh 78; Raj. 73; Ravishankar 72]

The required equations of the planes bisecting the angle between the given planes are

$$\frac{x + 2y + 2z - 9}{\sqrt{1 + 4 + 4}} = \pm \frac{4x - 3y + 12z + 13}{\sqrt{16 + 9 + 144}}$$

$$\text{or } 25x + 17y + 62z - 78 = 0 \text{ and } x + 35y - 10z - 156 = 0$$

Acute or Obtuse.

Now angle between the planes $x+2y+2z=9$ and the bisecting plane $x+35y-10z-156=0$ is given by

$$\cos \theta = \frac{1 \cdot 1 + 2(35) + 2(-10)}{\sqrt{(1^2 + 35^2 + 10^2)} \sqrt{(1^2 + 2^2 + 2^2)}} = \frac{51}{36 \times 3} = \frac{17}{36}$$

$$\therefore \tan \theta = \frac{\sqrt{(1007)}}{17} = \sqrt{\left(\frac{1007}{289}\right)} \text{ i.e. } > 1 : \therefore \theta > 45^\circ$$

Hence the plane $x-35y-10z-156=0$ bisects the obtuse angle between the planes and hence the other plane $25x+17y+62z-78=0$ bisects the acute angle.

With respect to origin.

Again writing the equations of the planes so that the constant terms in both are positive,

i.e. $-x-2y-2z+9=0$ and $4x-3y+12z+13=0$, then the plane

$$\frac{-x-2y-2z+9}{\sqrt{(1^2+2^2+2^2)}} = \pm \frac{4x-3y+12z+13}{\sqrt{(4^2+3^2+12^2)}}$$

i.e. $25x+17y+62z-78=0$ bisects the angle between the planes that contains the origin.

By the Combined Method.

The two given planes are

$$x+2y+2z-9=0 \text{ and } 4x-3y+12z+13=0.$$

Reducing them to the form so that the constant terms in both of them are +ive, we have $-x-2y-2z+9=0$ and

$$4x-3y+12z+13=0.$$

$$\therefore a_1a_2+b_1b_2+c_1c_2 = -1(4) - 2(-3) - 2(12) = -22;$$

i.e. -ive and hence the origin lies in the acute angle between the planes and the equation of the bisecting plane that bisects the angle, containing the origin by Method II is

$$25x+17y+62z-78=0$$

and since we have seen that origin lies in the acute angle, the bisecting plane bisects the obtuse angle between the planes.

Ex. 2. Find the plane that bisects the acute angle between the planes $2x-y+2z+3=0$ and $3x-2y+6z+8=0$.

[Pb. Pass 71; Rajputana 83]

The constant term in both the equations are positive. Again $a_1a_2+b_1b_2+c_1c_2=6+2+12=20$, i.e. +ive, hence origin lies in the obtuse angle between the planes. The equations of the bisecting planes are given by

$$\frac{2x-y+2z+3}{\sqrt{(2^2+1^2+2^2)}} = \pm \frac{3x-2y+6z+8}{\sqrt{(3^2+2^2+6^2)}}$$

or $7(2x - y + 2z + 3) = \pm 3(3x - 2y + 6z + 8)$

Taking +ive sign, we shall get the equation of the plane bisecting the angle between the planes in which the origin lies (in the question it lies in the obtuse angle). Hence the -ive sign gives the plane that bisects the angle between the planes in which the origin does not lie, i.e. acute.

Hence the required plane is

$$(14x - 7y + 14z + 21) = -(9x - 6y + 18z + 24)$$

$$23x - 13y + 32z + 45 = 0.$$

Ex. 3. Find the equation of the plane that bisects the angle between the planes

$$3x - 6y - 2z + 5 = 0 \text{ and } 4x - 12y + 3z - 3 = 0$$

which contains the origin.

Is this the plane that bisects the acute angle? [Utkal 82]

Ans. $67x + 162y + 47z + 44 = 0$, yes.

Ex. 4. Find the equation of the plane which bisects the acute angle between the planes

$$3x - 4y + 12z = 26 \text{ and } x + 2y - 2z = 9$$

Ans. $22x + 14y + 10z - 195 = 0$.

§ 13. Combined equation of two planes.

To find the condition that the general homogeneous equation of second degree in x, y and z may represent a pair of planes, and to find the angle between them and the condition that these two planes may be perpendicular to each other.

[Indore 78; Rohailkhand 82; Kanpur 77; Venkateshwara 67;

Karnatak 69; Allahabad 69; Gauhati Hons. 67; Raj. 84, 74;

Agra 67; Vikram 72; Pb. 71]

The general homogeneous equation of second degree in x, y and z is

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0. \quad \dots(1)$$

Let the equation of the two planes represented by (1) be

$$l_1x + m_1y + n_1z = 0 \text{ and } l_2x + m_2y + n_2z = 0.$$

There will be no constant term in the two equations representing the planes, for otherwise their product will not be homogeneous.

$$\therefore ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy \\ \equiv (l_1x + m_1y + n_1z)(l_2x + m_2y + n_2z).$$

Comparing the coefficients, we have

$$\left. \begin{aligned} l_1l_2 &= a, m_1m_2 = b, n_1n_2 = c, \\ m_1n_2 + m_2n_1 &= 2f, n_1l_2 + n_2l_1 = 2g, l_1m_2 + l_2m_1 = 2h \end{aligned} \right\} \quad \dots(2)$$

The required condition is obtained by eliminating l_1, m_1, n_1 and l_2, m_2, n_2 in the relation (2).

Now consider the product of two zero determinants

$$\begin{vmatrix} l_1 & l_2 & 0 \\ m_1 & m_2 & 0 \\ n_1 & n_2 & 0 \end{vmatrix} \times \begin{vmatrix} l_2 & l_1 & 0 \\ m_2 & m_1 & 0 \\ n_2 & n_1 & 0 \end{vmatrix} = 0$$

On multiplying the two determinants, we get

$$\begin{vmatrix} 2l_1l_2 & l_1m_2 + l_2m_1 & l_1n_2 + l_2n_1 \\ l_1m_2 + l_2m_1 & 2m_1m_2 & m_1n_2 + m_2n_1 \\ l_1n_2 + l_2n_1 & m_1n_2 + m_2n_1 & 2n_1n_2 \end{vmatrix}$$

On putting the values of $l_1l_2, m_1n_2 + m_2n_1$ etc. from (2), we get

$$\begin{vmatrix} 2a & 2h & 2g \\ 2h & 2b & 2f \\ 2g & 2f & 2c \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0 \quad \dots(1)$$

$$a(bc - f^2) - h(ch - fg) + g(hf - bg) = 0$$

or $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ is the required condition.

Angle between the planes.

[Indore 78; Bundelkhand 79; Kanpur 77; Raj. 74; Luck. 76]

Again if θ be the angle between the two planes, then

$$\tan \theta = \frac{\sqrt{[(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2 + (l_1m_2 - l_2m_1)^2]}}{l_1l_2 + m_1m_2 + n_1n_2}$$

Now

$$(m_1n_2 - m_2n_1)^2 = (m_1n_2 + m_2n_1)^2 - 4m_1m_2n_1n_2 = 4(f^2 - bc)$$

$$\text{or} \quad \theta = \tan^{-1} \frac{2\sqrt{(f^2 + g^2 + h^2 - ab - bc - ca)}}{a + b + c} \quad \dots(2)$$

In case the planes are perpendicular, then $\theta = \pi/2$.

$\therefore \tan \pi/2$ being ∞ , we get $a + b + c = 0$.

i.e. $\text{coeff. of } x^2 + \text{coeff. of } y^2 + \text{coeff. } z^2 = 0$.

[Raj. 84; Pb. 81]

Exercise 6

Ex. 1. If the equation

$$\phi(x, y, z) = ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0$$

represents a pair of planes, prove that the product of the distances of the two planes from (α, β, γ) is

$$\frac{\phi(\alpha, \beta, \gamma)}{\sqrt{(\Sigma a^2 + 4\Sigma f^2 - 2\Sigma bc)}} \quad [\text{Avadh 79}]$$

Let the equation of the two planes be

$$l_1x + m_1y + n_1z = 0 \text{ and } l_2x + m_2y + n_2z = 0$$

so that we have

$$(l_1x + m_1y + n_1z)(l_2x + m_2y + n_2z) \equiv \phi(x, y, z) \quad \dots(1)$$

$$l_1l_2 = a, m_1m_2 = b, n_1n_2 = c.$$

$$m_1n_2 + m_2n_1 = 2f, n_1l_2 + n_2l_1 = 2g, l_1m_2 + l_2m_1 = 2h.$$

If p_1 and p_2 be the perpendicular distances of the point (α, β, γ) from the two planes, then

$$p_1p_2 = \frac{l_1\alpha + m_1\beta + n_1\gamma}{\sqrt{l_1^2 + m_1^2 + n_1^2}} \cdot \frac{l_2\alpha + m_2\beta + n_2\gamma}{\sqrt{l_2^2 + m_2^2 + n_2^2}} \quad (\S 11. P. 62)$$

$$= \phi(\alpha, \beta, \gamma) / \sqrt{\{l_1^2l_2^2 + m_1^2m_2^2 + n_1^2n_2^2 + (l_1^2m_2^2 + m_1^2l_2^2) + (n_1^2l_2^2 + l_1^2n_2^2) + (m_1^2n_2^2 + m_2^2n_1^2)\}}$$

$$= \frac{\phi(\alpha, \beta, \gamma)}{\sqrt{[a^2 + b^2 + c^2 + \Sigma\{(l_1m_2 + l_2m_1)^2 - 2l_1l_2m_1m_2\}]}}$$

$$= \frac{\phi(\alpha, \beta, \gamma)}{\sqrt{[a^2 + b^2 + c^2 + 4h^2 - 2ab + 4g^2 - 2ac + 4f^2 - 2bc]}}$$

Ex. 2. Show that $\frac{a}{y-z} + \frac{b}{z-x} + \frac{c}{x-y} = 0$ represents a pair of planes. [Agra 78]

The given equation can be written as

$$a(z-x)(x-y) + b(y-z)(x-y) + c(y-z)(z-x) = 0$$

$$\text{or } ax^2 + by^2 + cz^2 - (b+c-a)yz - (c+a-b)zx - (a+b-c)xy = 0.$$

It will represent a pair of planes if

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0 \text{ or } \begin{vmatrix} a & -\frac{1}{2}(a+b-c) & -\frac{1}{2}(c+a-b) \\ -\frac{1}{2}(a+b-c) & b & -\frac{1}{2}(b+c-a) \\ -\frac{1}{2}(c+a-b) & -\frac{1}{2}(b+c-a) & c \end{vmatrix} = 0$$

If we add 2nd and third rows to first row, all its constituents become zero and hence the determinant vanishes. Therefore the given equation represents a pair of planes.

Ex. 3. Prove that the equation

$$2x^2 - 6y^2 - 12z^2 + 18yz + 2zx + xy = 0$$

represents a pair of planes, and find the angle between them.

[Indore 79, 71; L.N.M. 82; Jiwaji 66; Rajputana 67;

Garhwal 78; Madurai 82; Ranchi 80]

Put down the numerical values of a, b, c, f, g, h in result (1) and (2) of P. 65.

Alternative Method.

The terms of x and y i.e. $2x^2 + xy - 6y^2$ can be factorized as $(x+2y)(2x-3y)$. Now suppose that the factors of the given equation are $(x+2y+pz)(2x-3y+qz)$.

Comparing the coefficients of remaining terms of zx, yz and z^2 we get $2p+q=2, -3p+2q=18$ and $pq=-12$.

Solving the first two we get $p=-2, q=6$ and these values satisfy the third relation $pq=-12$. Therefore the given equation represents two planes whose equations are

$$x+2y-2z=0, 2x-3y+6z=0.$$

If θ be the angle between the normals to the planes whose D.R.'s are $1, 2, -2; 2, -3, 6$.

$$\text{Then } \cos \theta = \pm \frac{1 \cdot 2 + 2(-3) - 2(6)}{\sqrt{(1+4+4)} \sqrt{(4+9+36)}} = \pm \frac{16}{21}.$$

$$\theta = \cos^{-1} \left(\frac{16}{21} \right). \text{ [Acute angle].}$$

Ex. 4. Prove that the equation

$$2x^2 + 6y^2 - 12z^2 + 6yz + 2zx + 7xy = 0$$

represents a pair of planes and find the angle between them.

Ans. $\cos^{-1} \frac{4}{5}$.

§ 14. Projection on a plane.

Just as we have defined the projection of a point and projection of the segment of a line on a line (§ P. 23), similarly the projection of area A' enclosed by the curve $PQR...$ on a plane is the area A' enclosed by the curve $P'Q'R'...$ where $P', Q', R', ...$ are feet of the perpendiculars from $PQR...$ on the plane of projection and we shall assume here the result of Pure Solid Geometry that the area $A' = A \cos \theta$, where θ is the angle between the plane of the area A and the plane of projection.

Two Important Theorems.

1. If A_x, A_y and A_z be the projections of an area A on the co-ordinate planes, viz, yz, zx and xy planes respectively, then

$$A^2 = A_x^2 + A_y^2 + A_z^2$$

A_x is equal to the projection of area A on the yz -plane.

$\therefore A_x = A \cos \alpha$ where α is the angle between the plane of area A and yz -plane, i.e. between their normals. Let us suppose that l, m, n are the direction-cosines of the normal to the plane of area A . Then $l = \cos$ of the angle between the normal and x -axis (which is normal to the yz -plane) $= \cos \alpha$.

Note. The above result is identical with the corresponding result of plane geometry.

$\therefore A_x = Al$. Similarly $A_y = Am$ and $A_z = An$.

Squaring and adding, we get

$$A_x^2 + A_y^2 + A_z^2 = A^2(l^2 + m^2 + n^2) = A^2$$

II. To prove that projection of any plane area A on a given plane π is equal to the sum of the projections of A_x , A_y and A_z on the given plane π where A_x , A_y and A_z are the projections of the area A on the co-ordinate planes.

Let l_1, m_1, n_1 be the direction-cosines of the normal to the plane of the area A and l_2, m_2, n_2 be the d.c.'s of the normal to the plane of projection π so that if θ be the angle between the two planes then $\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$ (1)

Now if A' be the projection of area A on the plane π , then $A' = A \cos \theta = A (l_1 l_2 + m_1 m_2 + n_1 n_2)$.

Now by definition $A_x = Al_1$, $A_y = Am_1$, $A_z = An_1$. (Theorem 1)

$$\therefore A' = A_x l_2 + A_y m_2 + A_z n_2. \quad \dots (2)$$

Now the projection of area A_x (which is the yz -plane on the plane π , d.c.'s of whose normals are l_2, m_2, n_2 is $A_x l_2$.

Hence from (2), we get

$A' =$ sum of the projections of the areas A_x , A_y and A_z on the given plane π .

Cor. 1. To find the area of the triangle, the co-ordinates of whose vertices are given.

Now if (a, b, c) be the co-ordinates of a point P , then the co-ordinates of the point P_x which is its projection on the yz -plane are $(0, b, c)$ and those of p_y which is its projection on the zx -plane are $(a, 0, c)$ and those of p_z will be $(a, b, 0)$.

Then we shall project the three given vertices on the xy -plane and find the plane area A_x which will be projection of the area of the triangle on the yz -plane. By using the formula of the plane geometry.

$$\Delta = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Similarly we shall find the values of A_y and A_z and then by theorem 1, area $A = \sqrt{A_x^2 + A_y^2 + A_z^2}$.

Exercise 7

Ex. 1. Find the area of the triangle, the co-ordinates of whose vertices are $(1, 2, 3)$, $(-2, 1, -4)$, $(3, 4, -2)$. [Agra 79]

Co-ordinates of the points of projection in the yz -plane of the three vertices are $(0, 2, 3)$, $(0, 1, -4)$, $(0, 4, -2)$ or in plane geometry these points are $(2, 3)$, $(1, -4)$, and $(4, -2)$ and the plane area A_x of the triangle formed by these points is

$$\frac{1}{2} \begin{vmatrix} 2 & 3 & 1 \\ 1 & -4 & 1 \\ 4 & -2 & 1 \end{vmatrix} = \frac{19}{2}$$

Similarly, $A_y = \frac{13}{2}$ and $A_z = -2$.

$$A = \sqrt{(A_x^2 + A_y^2 + A_z^2)} = \frac{1}{2} \sqrt{(1218)}.$$

Ex. 2. A plane makes intercepts $OA=a$, $OB=b$ and $OC=c$ respectively on the axes of co-ordinates. Find the area of the triangle ABC . [Kanpur 77 ; Bundelkhand 79 ; Allahabad 67]

A_x = area of the plane triangle formed by the three points which are the projections of the vertices on the yz -plane.

Clearly the points B and C are in the yz -plane and the projection of the point $A(a, 0, 0)$ is $(0, 0, 0)$;

$\therefore A_x$ = area of $\triangle OBC$, which is right angled at O .

$$\therefore A_x = \frac{1}{2} bc.$$

Similarly, $A_y = \frac{1}{2} ca$, $A_z = \frac{1}{2} ab$.

$$\therefore A = \frac{1}{2} \sqrt{(b^2c^2 + c^2a^2 + a^2b^2)}.$$

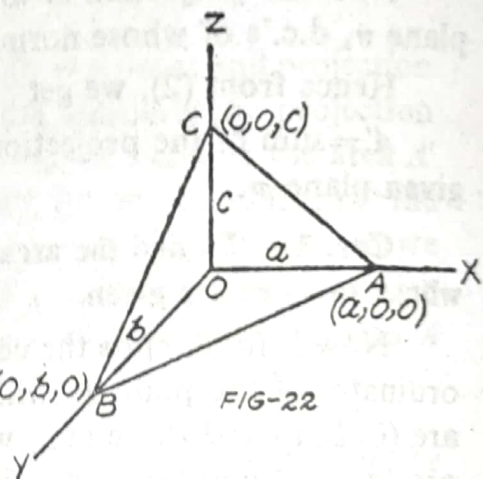
Ex. 3. From a point $P(x', y', z')$ a plane is drawn at right angles to OP to meet the co-ordinate axes at A , B and C . Prove that area of the triangle ABC is $r^5/2x'y'z'$ where $r=OP$.

[Rohailkhand 83 ; Vikram 84 ; Ravishankar 81 ; Agra 81 ; Kanpur 80 ; Utkal Hons. 67 ; Pb. 77 ; Rajputana 81, 65 ; Rewa 71]

The equation of the plane through P at right angles to OP is

$$x'(x-x') + y'(y-y') + z'(z-z') = 0 \quad (\text{Q. 5 p. 44})$$

$$\text{or } xx' + yy' + zz' = x'^2 + y'^2 + z'^2 = r^2 \quad \therefore OP = r$$



The points A, B, C where the plane meets the axes of co-ordinates are

$$\left(\frac{r^2}{x'}, 0, 0\right), \left(0, \frac{r^2}{y'}, 0\right), \left(0, 0, \frac{r^2}{z'}\right)$$

If A_x be the projection of the $\triangle ABC$ on the yz -plane then we know $A_x = \triangle ABC \cdot l$, ... (1)

where l, m, n , are d.c.'s of the normal to the plane of area ABC .

Now the projection of $\triangle ABC$ on the yz -plane is clearly right-angled $\triangle OBC$ whose area as in Ex. 2 is

$$A_x = \frac{1}{2} OB \cdot OC = \frac{1}{2} \frac{r^2}{x'} \cdot \frac{r^2}{y'} = \frac{1}{2} \cdot \frac{r^4}{y'z'}$$

Now the direction-ratios of normal to ABC being x', y', z' , the actual direction-cosines are

$$\frac{x'}{\sqrt{(x'^2 + y'^2 + z'^2)}}, \frac{y'}{\sqrt{(x'^2 + y'^2 + z'^2)}}, \frac{z'}{\sqrt{(x'^2 + y'^2 + z'^2)}}$$

i.e. $\left(\frac{x'}{r}, \frac{y'}{r}, \frac{z'}{r}\right)$;

$\therefore l = \frac{x'}{r}$. Putting the values of A_x and l in (1), we get

$$\frac{1}{2} \cdot \frac{r^4}{y'z'} = \triangle ABC \frac{x'}{r} \quad \therefore \triangle ABC = \frac{1}{2} \cdot \frac{r^5}{x'y'z'}$$

The Straight Line

§ 1. General Equation of Straight Line.

Any two equations of first degree in x, y and z taken together represent a line. [Punjab 80]

We have already proved in Chapter III that every equation of first degree in x, y and z represents a plane.

Now if we take two equations of first degree together i.e. $a_1x + b_1y + c_1z + d_1 = 0, a_2x + b_2y + c_2z + d_2 = 0 \dots (1)$ then any point which satisfies their equations individually will satisfy both of them as well. Hence these two equations taken together represent the curve of intersection of the two given planes and we know that two planes intersect in a straight line.

Then equations (1) represent the equation of a straight line which is the line of intersection of the two planes.

§ 2. Symmetrical form of the equations of a line. [Distance Form]. [Calicut 74 ; Punjab 50]

To find the equations of a straight line that passes through a given point (x_1, y_1, z_1) and whose direction cosines are l, m, n .

Let A be the given point (x_1, y_1, z_1) on the line. Choose any point $P(x, y, z)$ on it at a distance r from A . Now if A' and P' be the feet of the perpendiculars from A and P on the axis of x , then $A'P' = x - x_1$ measures the projection of AP on the axis of x .

$\therefore A'P' = AP \cos \alpha$, where α is the angle between the two lines AP and $A'P'$, i.e. between AP and x -axis, or $x - x_1 = rl$.

[§ 6, P. 18]

Similarly projecting on y -axis and z -axis, we shall get

$$y - y_1 = rm, z - z_1 = rn,$$

$$\therefore \frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r$$